

Deterministic Optimization

Discrete Optimization:
Modeling w/ Binary Variables

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Nonconvex functions

Modeling Nonconvex Functions

Learning Objectives

- Figure out how to model nonconvex functions using mixed integer **linear** constraints

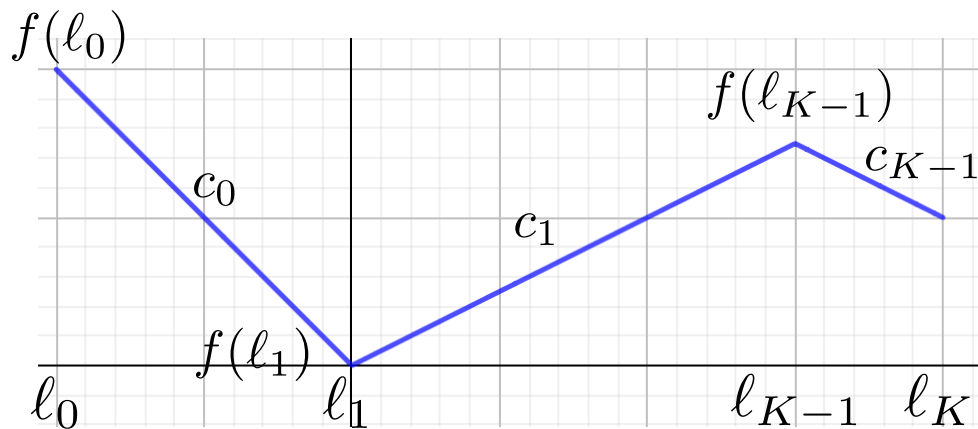
Problem Class

- Bounded variables
- Separable objective function, i.e. sum of univariate functions
- Piecewise linear
- Nonconvex and nonlinear
- Continuous

$$\begin{array}{ll}\min & \sum_{j=1}^n f_j(x_j) \\ \text{s.t.} & \mathbf{Ax} \geq \mathbf{b} \\ & 0 \leq x_j \leq U_j \quad j = 1, \dots, n\end{array}$$

Piecewise Linear Function

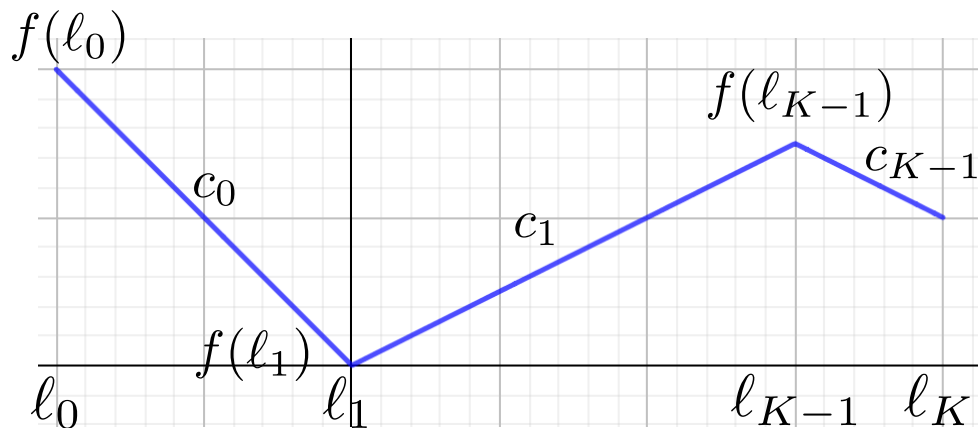
$$f(x) = f(\ell_k) + c_k(x - \ell_k), \quad \ell_k \leq x \leq \ell_{k+1}, \quad k = 0, \dots, K-1$$



Piecewise Linear Function

function value at breakpoint slope breakpoint

$$f(x) = f(\ell_k) + c_k(x - \ell_k), \quad \ell_k \leq x \leq \ell_{k+1}, \quad k = 0, \dots, K-1$$



Modeling Approach 1

Introduce new variables $u_0, \dots, u_{K-1}$ and $y_0, \dots, y_{K-1}$.

Introduce the following constraints:

$$\begin{aligned}x &= \sum_{k=0}^{K-1} u_k \\ \ell_k y_k &\leq u_k \leq \ell_{k+1} y_k \quad k = 0, \dots, K-1 \\ \sum_{k=0}^{K-1} y_k &= 1 \\ y_k &\in \{0, 1\} \quad k = 0, \dots, K-1\end{aligned}$$

Replace $f(x)$ by $\sum_{k=0}^{K-1} [f(\ell_k) + c_k(u_k - \ell_k)]$

A Wrong Approach

Introduce new variables $y_0, \dots, y_{K-1}$.

Introduce the following constraints:

$$\ell_k y_k \leq x \leq \ell_{k+1} y_k \quad k = 0, \dots, K - 1$$

$$\sum_{k=0}^{K-1} y_k = 1$$

$$y_k \in \{0, 1\} \quad k = 0, \dots, K - 1$$

Replace $f(x)$ by $\sum_{k=0}^{K-1} [f(\ell_k) + c_k(x - \ell_k)] y_k$

Correct formulation but is **nonlinear** !!!

Example

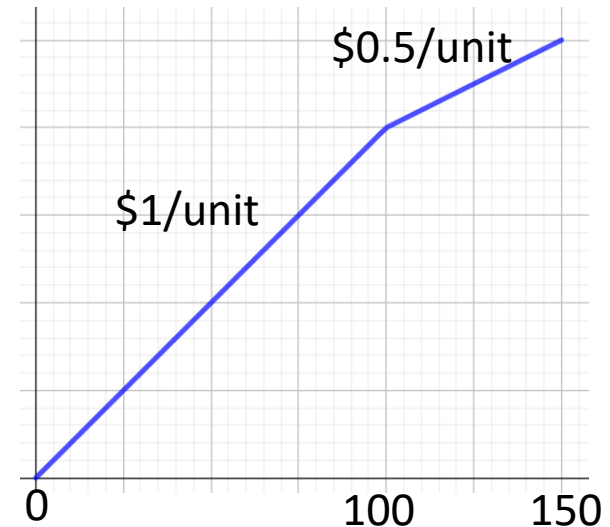
- Decide how much to buy from a set of different vendors (offering quantity discounts) to satisfy a total need

$$\min \sum_{j=1}^n f_j(x_j)$$

$$\text{s.t. } \mathbf{Ax} \geq \mathbf{b}$$

$$0 \leq x_j \leq U_j \quad j = 1, \dots, n$$

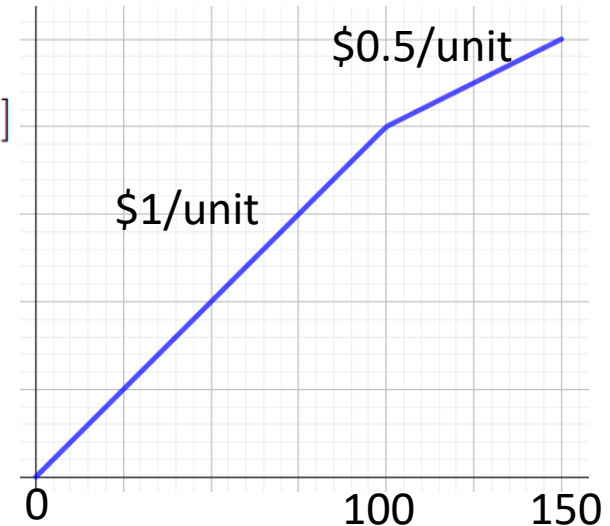
$$f_j(x_j) = \begin{cases} p_j x_j & 0 \leq x_j \leq L_j \\ p_j L_j + q_j (x_j - L_j) & L_j \leq x_j \leq U_j \end{cases}$$



Example

- Decide how much to buy from a set of different vendors (offering quantity discounts) to satisfy a total need

$$\begin{array}{ll}\min & \sum_{j=1}^n [p_j u_j + p_j L_j y_j + q_j (v_j - L_j) + q_j L_j (1 - y_j)] \\ \text{s.t.} & Ax \geq b \\ & 0 \leq x_j \leq U_j, \quad j = 1, \dots, n \\ & x_j = u_j + v_j, \quad j = 1, \dots, n \\ & 0 \leq u_j \leq L_j(1 - y_j), \quad j = 1, \dots, n \\ & L_j y_j \leq v_j \leq U_j y_j, \quad j = 1, \dots, n \\ & y_j \in \{0, 1\}, \quad j = 1, \dots, n\end{array}$$



Modeling Approach 2

Introduce new variables $\lambda_0, \dots, \lambda_K$ and $y_0, \dots, y_{K-1}$

Introduce the following constraints:

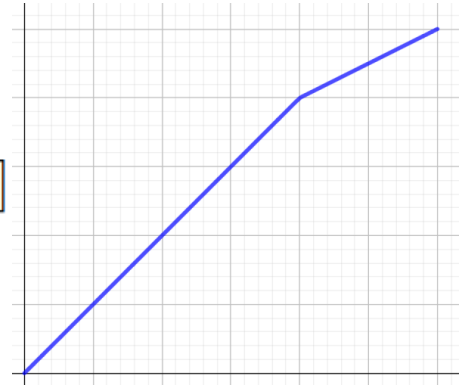
$$\begin{aligned} x &= \sum_{k=0}^K \lambda_k \ell_k, \quad \sum_{k=0}^K \lambda_k = 1, \quad \sum_{k=0}^{K-1} y_k = 1 \\ y_k &\in \{0, 1\}, \quad k = 0, \dots, K-1, \quad \lambda_k \geq 0, \quad k = 0, \dots, K \\ \lambda_0 &\leq y_0, \quad \lambda_k \leq y_{k-1} + y_k, \quad k = 1, \dots, K-1, \quad \lambda_K \leq y_{K-1} \end{aligned}$$

Replace $f(x)$ by $\sum_{k=0}^K f(\ell_k) \lambda_k$

Example

- Decide how much to buy from a set of different vendors (offering quantity discounts) to satisfy a total need

$$\begin{aligned} \min \quad & \sum_{j=1}^n [(0)\lambda_{j,1} + (p_j L_j)\lambda_{j,2} + (p_j L_j + q_j(U_j - L_j))\lambda_{j,3}] \\ \text{s.t.} \quad & Ax \geq b, \quad 0 \leq x_j \leq U_j, \quad j = 1, \dots, n \\ & x_j = (0)\lambda_{j,1} + (L_j)\lambda_{j,2} + (U_j)\lambda_{j,3}, \quad j = 1, \dots, n \\ & \lambda_{j,1} + \lambda_{j,2} + \lambda_{j,3} = 1, \quad j = 1, \dots, n, \quad \lambda \geq 0 \\ & y_{j,1} + y_{j,2} = 1, \quad j = 1, \dots, n, \quad y \in \{0, 1\} \\ & \lambda_{j,1} \leq y_{j,1}, \quad \lambda_{j,2} \leq y_{j,1} + y_{j,2}, \quad \lambda_{j,3} \leq y_{j,2}, \quad j = 1, \dots, n \end{aligned}$$



Summary

- We learned two ways to model piece-wise linear nonconvex functions by introducing new continuous and binary variables
- It is very important to make sure the resulting formulations are **linear**